



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 11/01/2024*

PROBLEM - We observe the following two data samples:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \mathbf{S} + \mathbf{W} = \begin{bmatrix} A \cos \theta \\ A \sin \theta \end{bmatrix} + \begin{bmatrix} W_1 \\ W_2 \end{bmatrix}$$

where the phase θ is uniformly distributed in $[-\pi, \pi)$ and the amplitude A is Rayleigh distributed,

$$f_A(a) = \frac{2a}{\sigma_A^2} e^{-\frac{a^2}{\sigma_A^2}} u(a), \quad E\{A\} = \sqrt{\frac{\pi}{4}} \sigma_A, \quad E\{A^2\} = \sigma_A^2.$$

A and θ are mutually independent and are both independent on the noise samples W_1 and W_2 . The noise samples W_1 and W_2 are identically distributed correlated Gaussian random variables, with zero mean, variance σ_w^2 , and correlation coefficient ρ .

Derive:

(1) the signal to noise ratio (SNR): $\gamma = \frac{E\{\|\mathbf{S}\|^2\}}{E\{\|\mathbf{W}\|^2\}};$

(2) the LMMSE estimator of the phase θ and express it as a function of σ_A , γ and ρ ;

(3) the bias and mean square error (MSE) of the LMMSE estimator of θ and express it as a function of γ and ρ ; then plot the MSE as a function of ρ ;

(4) Discuss the consistency of the LMMSE estimator and discuss and compare the two cases $\rho=0$ and $\rho=1$.

Hint: $\frac{1}{2\pi} \int_{-\pi}^{\pi} z \cos(z) dz = 0, \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} z \sin(z) dz = 1$



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Solution

(1) The signal to noise ratio (SNR): $\gamma = \frac{E\{\|\mathbf{S}\|^2\}}{E\{\|\mathbf{W}\|^2\}}$

$$E\{\|\mathbf{S}\|^2\} = E\{(A \cos \theta)^2 + (A \sin \theta)^2\} = E\{A^2\} = \sigma_A^2$$

$$E\{\|\mathbf{W}\|^2\} = E\{W_1^2 + W_2^2\} = 2\sigma_w^2$$

$$\Rightarrow \gamma = \frac{E\{\|\mathbf{S}\|^2\}}{E\{\|\mathbf{W}\|^2\}} = \frac{\sigma_A^2}{2\sigma_w^2}$$

(2) By applying the **Orthogonality principle**, we find that the LMMSE estimator is given by:

$$\hat{\theta}_{LMMSE} = \mathbf{a}^T \mathbf{X} + b = \eta_\theta + \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} (\mathbf{X} - \boldsymbol{\eta}_X)$$

where

$$\eta_\theta = E\{\theta\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta d\theta = 0$$

$$\boldsymbol{\eta}_X = \begin{bmatrix} E\{X_1\} \\ E\{X_2\} \end{bmatrix} = \begin{bmatrix} E\{A \cos \theta\} + E\{W\} \\ E\{A \sin \theta\} + E\{W\} \end{bmatrix} = \begin{bmatrix} E\{A\} E\{\cos \theta\} + E\{W\} \\ E\{A\} E\{\sin \theta\} + E\{W\} \end{bmatrix} = \sqrt{\frac{\pi}{4}} \sigma_A \begin{bmatrix} \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \theta d\theta \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin \theta d\theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Since the parameter θ and the data $\mathbf{X}$ are zero mean, we have that:

$$b = \eta_\theta - \mathbf{a}^T \boldsymbol{\eta}_X = 0 \Rightarrow \hat{\theta}_{LMMSE} = \mathbf{a}^T \mathbf{X} = \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{X}$$

where the coefficient vector $\mathbf{a}$ is derived by applying the Orthogonality Principle: $\mathbf{a} = \mathbf{C}_X^{-1} \mathbf{c}_{X\theta}$



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$$\mathbf{c}_{X\theta} = \begin{bmatrix} \text{cov}\{X_1, \theta\} \\ \text{cov}\{X_2, \theta\} \end{bmatrix} = \begin{bmatrix} E\{X_1\theta\} \\ E\{X_2\theta\} \end{bmatrix}, \quad \mathbf{C}_X = \begin{bmatrix} \text{var}\{X_1\} & \text{cov}\{X_1, X_2\} \\ \text{cov}\{X_1, X_2\} & \text{var}\{X_2\} \end{bmatrix} = \begin{bmatrix} E\{X_1^2\} & E\{X_1X_2\} \\ E\{X_1X_2\} & E\{X_2^2\} \end{bmatrix}$$

$$E\{X_1\theta\} = E\{A\theta \cos \theta + W\theta\} = E\{A\} E\{\theta \cos \theta\} = 0$$

$$E\{X_2\theta\} = E\{A\theta \sin \theta + W\theta\} = E\{A\} E\{\theta \sin \theta\} = \sqrt{\frac{\pi}{4}} \sigma_A$$

$$\begin{aligned} E\{X_1^2\} &= E\{(A \cos \theta + W_1)(A \cos \theta + W_1)\} \\ &= E\{A^2\} E\{\cos^2 \theta\} + 2E\{A\} E\{\cos \theta\} E\{W_1\} + E\{W_1^2\} \\ &= \frac{\sigma_A^2}{2} (1 + E\{\cos(2\theta)\}) + \sigma_w^2 = \frac{\sigma_A^2}{2} + \sigma_w^2 \end{aligned}$$

$$\begin{aligned} E\{X_2^2\} &= E\{(A \sin \theta + W_2)(A \sin \theta + W_2)\} \\ &= E\{A^2\} E\{\sin^2 \theta\} + 2E\{A\} E\{\sin \theta\} E\{W_2\} + E\{W_2^2\} \\ &= \frac{\sigma_A^2}{2} (1 - E\{\cos(2\theta)\}) + \sigma_w^2 = \frac{\sigma_A^2}{2} + \sigma_w^2 \end{aligned}$$

$$\begin{aligned} E\{X_1X_2\} &= E\{(A \cos \theta + W_1)(A \sin \theta + W_2)\} \\ &= E\{A^2\} E\{\cos \theta \sin \theta\} + E\{A\} E\{\cos \theta\} E\{W_2\} \\ &\quad + E\{A\} E\{\sin \theta\} E\{W_1\} + E\{W_1W_2\} \\ &= \frac{\sigma_A^2}{2} E\{\sin(2\theta)\} + \rho\sigma_w^2 = \rho\sigma_w^2 \end{aligned}$$

$$\text{where } E\{A^2\} = \sigma_A^2, \quad E\{W_1W_2\} = \rho\sigma_w^2, \quad E\{W_1\} = E\{W_2\} = 0$$

Hence, the LMMSE estimator is given by:

$$\hat{\theta}_{LMMSE} = \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{X} = \begin{bmatrix} 0 & \sqrt{\frac{\pi}{4}} \sigma_A \end{bmatrix} \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_w^2 & \rho\sigma_w^2 \\ \rho\sigma_w^2 & \frac{\sigma_A^2}{2} + \sigma_w^2 \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$



$$\begin{aligned}\hat{\theta}_{LMMSE} &= \begin{bmatrix} 0 & \sqrt{\frac{\pi}{4}} \sigma_A \end{bmatrix} \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_w^2 & \rho \sigma_w^2 \\ \rho \sigma_w^2 & \frac{\sigma_A^2}{2} + \sigma_w^2 \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \\ &= \sqrt{\frac{\pi}{4}} \sigma_A \begin{bmatrix} 0 & 1 \end{bmatrix} (\sigma_w^2)^{-1} \begin{bmatrix} 1+\gamma & \rho \\ \rho & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{\sqrt{\pi} \sigma_A}{2\sigma_w^2} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & \rho \\ \rho & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}\end{aligned}$$

where $\gamma \triangleq \frac{\sigma_A^2}{2\sigma_w^2}$ is the Signal to Noise Ratio (SNR)

$$\begin{aligned}\hat{\theta}_{LMMSE} &= \frac{\sqrt{\pi} \sigma_A}{2\sigma_w^2} ((1+\gamma)^2 - \rho^2)^{-1} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & -\rho \\ -\rho & 1+\gamma \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \\ &= \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\sigma_A^2}{2\sigma_w^2} \cdot \frac{1}{(1+\gamma)^2 - \rho^2} \begin{bmatrix} -\rho & 1+\gamma \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma}{(1+\gamma)^2 - \rho^2} (-\rho X_1 + (1+\gamma) X_2) \\ &= \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2} \left(X_2 - \frac{\rho}{1+\gamma} X_1 \right)\end{aligned}$$

In summary:

$$\hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2} \left(X_2 - \frac{\rho}{1+\gamma} X_1 \right)$$

It is worth observing that even if X_1 and θ are uncorrelated, the LMMSE estimator makes use of the observed sample X_1 , since it is correlated with X_2 . In practice, the role of X_1 is to estimate the noise sample W_2 and remove it from the observed sample X_2 , that is used for the estimate of θ .

$$\hat{W}_{2LMMSE} = \frac{\rho}{1+\gamma} X_1 \rightarrow X_2 - \frac{\rho}{1+\gamma} X_1 = X_2 - \hat{W}_{2LMMSE}$$

In fact, since X_1 and W_2 are zero mean, we have:



$$\begin{aligned}\hat{W}_{2LMMSE} &= \rho_{W_2X_1} \frac{\sigma_W}{\sigma_{X_1}} X_1 = \frac{\text{cov}\{W_2, X_1\}}{\sigma_{X_1}^2} X_1 = \frac{E\{X_1 W_2\}}{\frac{\sigma_A^2}{2} + \sigma_W^2} X_1 = \frac{E\{(A \cos \theta + W_1) W_2\}}{\frac{\sigma_A^2}{2} + \sigma_W^2} X_1 \\ &= \frac{E\{W_1 W_2\}}{\frac{\sigma_A^2}{2} + \sigma_W^2} X_1 = \frac{\rho \sigma_W^2}{\frac{\sigma_A^2}{2} + \sigma_W^2} X_1 = \frac{\rho}{\frac{\sigma_A^2}{2\sigma_W^2} + 1} X_1 = \frac{\rho}{1 + \gamma} X_1\end{aligned}$$

$$\hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2} \left(X_2 - \frac{\rho}{1+\gamma} X_1 \right) = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2} (X_2 - \hat{W}_{2LMMSE})$$

(3) The **bias** and the **mean square error (MSE)** of the LMMSE estimator are given by:

$$\text{bias}\{\hat{\theta}_{LMMSE}\} = E\{\theta - \hat{\theta}_{LMMSE}\} = E\{\theta\} - E\{\mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{X}\} = E\{\theta\} - \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} E\{\mathbf{X}\} = 0$$

Hence, the estimator is unbiased.

$$MSE\{\hat{\theta}_{LMMSE}\} = \sigma_\varepsilon^2 = \sigma_\theta^2 - \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{c}_{X\theta}$$

$$\sigma_\theta^2 = E\{\theta^2\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \frac{\pi^2}{3}$$

$$\begin{aligned}\sigma_\varepsilon^2 &= \sigma_\theta^2 - \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{c}_{X\theta} = \frac{\pi^2}{3} - \begin{bmatrix} 0 & \sqrt{\frac{\pi}{4}} \sigma_A \end{bmatrix} \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_W^2 & \rho \sigma_W^2 \\ \rho \sigma_W^2 & \frac{\sigma_A^2}{2} + \sigma_W^2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \sqrt{\frac{\pi}{4}} \sigma_A \end{bmatrix} \\ &= \frac{\pi^2}{3} - \frac{\pi \sigma_A^2}{4} \cdot \frac{1}{\sigma_W^2} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & \rho \\ \rho & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \frac{\pi^2}{3} - \frac{\pi}{2} \cdot \frac{\gamma}{(1+\gamma)^2 - \rho^2} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & -\rho \\ -\rho & 1+\gamma \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \frac{\pi^2}{3} - \frac{\pi}{2} \cdot \frac{\gamma}{(1+\gamma)^2 - \rho^2} \begin{bmatrix} -\rho & 1+\gamma \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{\pi^2}{3} - \frac{\pi}{2} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2}\end{aligned}$$

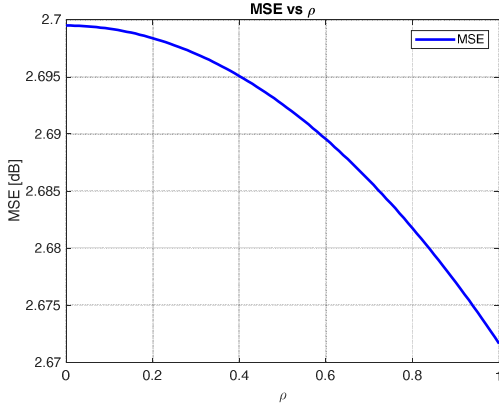
$$\text{In summary: } MSE\{\hat{\theta}_{LMMSE}\} = \frac{\pi^2}{3} - \frac{\pi}{2} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2}$$



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Plot of MSE [dB] versus ρ for $\gamma=10$:



The MSE decreases when ρ increases from 0 to 1.

(4) Consistency of the LMMSE estimator and the two cases $\rho=0$ and $\rho=1$.

$$\lim_{\gamma \rightarrow \infty} MSE\{\hat{\theta}_{LMMSE}\} = \frac{\pi^2}{3} - \frac{\pi}{2} \neq 0 \Rightarrow \text{the estimator is not consistent}$$

This is because there is a non-linear relationship between the observed data samples and the parameter to be estimated.

$$\rho=0: \quad \hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma}{1+\gamma} X_2, \quad MSE\{\hat{\theta}_{LMMSE}\} = \frac{\pi^2}{3} - \frac{\pi}{2} \cdot \frac{\gamma}{1+\gamma}$$

$$\rho=1: \quad \hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{1+\gamma}{2+\gamma} \left(X_2 - \frac{1}{1+\gamma} X_1 \right), \quad MSE\{\hat{\theta}_{LMMSE}\} = \frac{\pi^2}{3} - \frac{\pi}{2} \cdot \frac{1+\gamma}{2+\gamma}$$

When $\rho=0$, X_1 is not used by the LMMSE estimator, since it is uncorrelated both with θ and W_2 . Hence, it is useless for the linear estimate of θ . As a consequence the MSE in the case $\rho=0$ is larger than in the case $\rho=1$.